BrightCart – Comprehensive Profitability, Returns, and Marketing ROI Analysis

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1. Executive Overview

BrightCart is generating revenue at scale: 2,000 orders processed across eight product categories and four sales channels over two years, yielding ₹236,318 in net revenue at a 23.84% blended profit margin. The business is not failing. The issue is that this blended margin masks meaningful variation, and that variation is where the profitability opportunity lies.

Three structural issues are compressing margins. First, the product mix is suboptimal: Electronics and Toys generate 31% and 26% margins, respectively, yet represent only 26% of total order volume, while Books and Beauty generate 12% and 17% margins and incur disproportionate shipping costs relative to their price points. Second, the channel mix is working against unit economics: Marketplace and Social Commerce channels collectively processed 616 orders while paying ₹9,894 in platform fees that Website and Mobile App do not incur.

Social Commerce also carries the highest return rates in the business, making it the most expensive channel to operate on a fully loaded basis. Third, the marketing budget is misaligned with platform performance: Email Marketing delivers a 4.81 ROAS at a ₹17.86 cost per acquisition, while TikTok Ads delivers

24.02 ROAS at ₹3.06 CPA, yet TikTok receives the smallest budget allocation of any paid platform.

The core finding is that BrightCart's declining margins are a cost structure and allocation problem, not a demand problem. Demand is healthy. The opportunity is to extract more profit from existing order volume by shifting the product mix, migrating customers toward owned channels, reducing return leakage, and reallocating marketing spend toward demonstrably efficient platforms.

Five prioritized actions have been identified that collectively represent ₹26,700–₹40,900 in additional annual profit, a 47–73% improvement over the current ₹56,334 profit base, without acquiring a single new customer. The highest-impact quick win is marketing budget reallocation, executable within 30 days. The highest long-term lever is product mix optimization toward Electronics and Toys, executable within 90 days through targeted promotions and inventory prioritization.

2.Business Problem & Analytical Approach

2.1 Business Problem

BrightCart has experienced strong revenue growth over the past two years, generating more than **₹1M in gross revenue across multiple product categories and sales channels**. Despite this growth, the company has observed **declining net profit margins**, indicating that increased sales volume is not translating into proportional profitability.

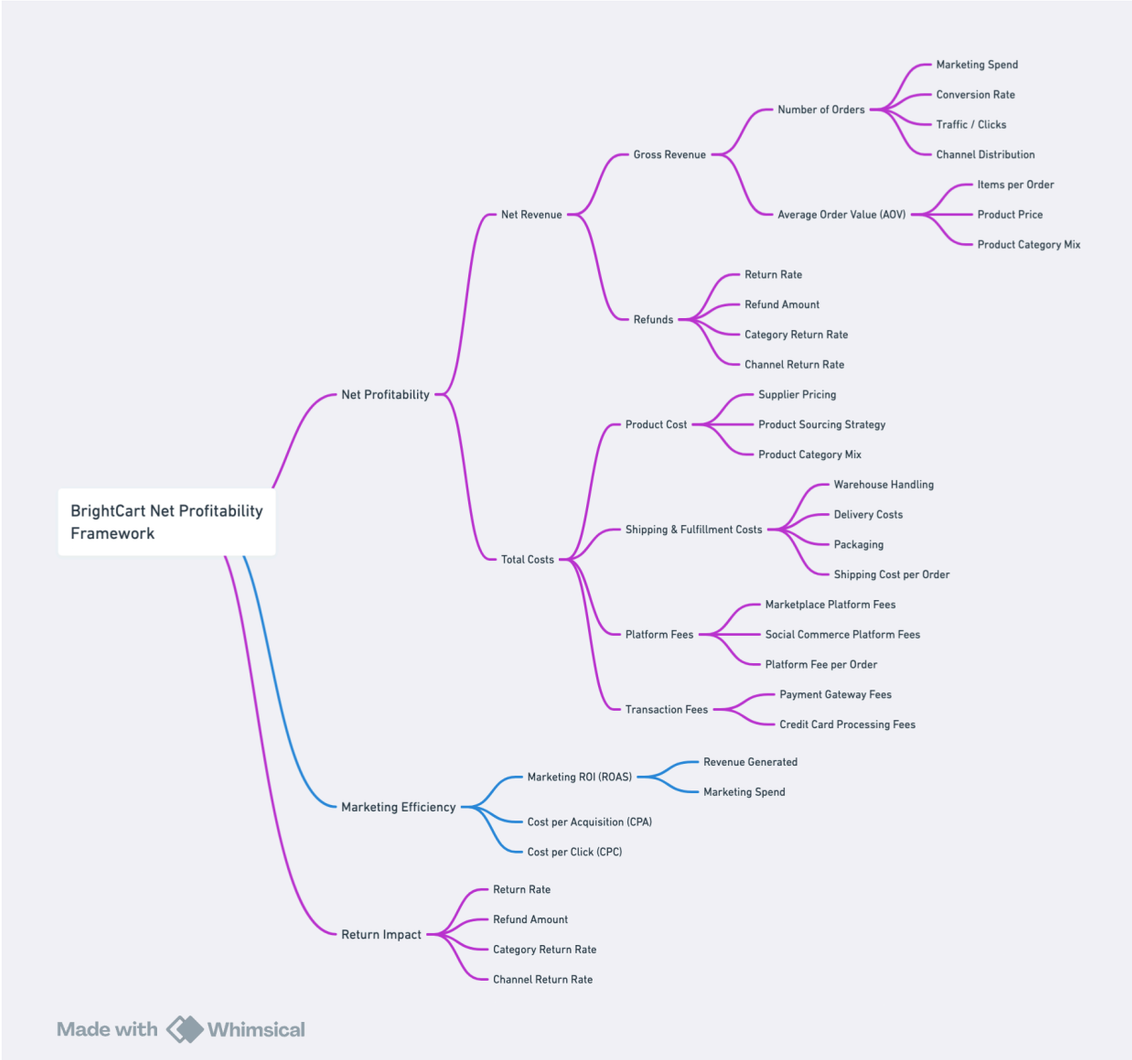
For an e-commerce business operating across multiple channels and marketing platforms, profitability is influenced by several interconnected factors including **product costs, fulfillment expenses, return rates, platform commissions, and marketing efficiency**. Without a clear understanding of how these factors interact, revenue growth alone may mask underlying inefficiencies that erode margins.

To address this challenge, BrightCart's leadership sought a comprehensive analysis to answer several critical business questions:

- Which **product categories generate sustainable profit margins**?
- How does **profitability vary across sales channels** such as Website, Mobile App, Marketplace, and Social Commerce?
- Are **product returns contributing significantly to margin erosion**?
- Which **marketing platforms generate the highest return on advertising investment**?
- Where can BrightCart **reduce marketing spend by 20% without significantly impacting revenue performance**?

Answering these questions requires analyzing BrightCart's **unit economics, operational cost structure, and marketing efficiency** in a structured and data-driven manner.

KPI Tree:



2.2 Analytical Approach

To investigate these questions, a structured analytical approach was applied using **transactional data, product cost data, and marketing performance data**.

The analysis was conducted in several stages.

1. Data Understanding and Validation

The first step involved understanding the structure of the available datasets and validating key financial metrics. Cost and profit calculations were verified to ensure that derived metrics such as **total costs and profit margins accurately reflected their underlying components**.

This step ensured that the analysis was based on **reliable and consistent financial data**.

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2. Unit Economics and Profitability Analysis

Using SQL-based analysis, order-level transactional data was examined to evaluate **revenue, costs, and profit margins across product categories**. This

allowed identification of which categories contribute the most to profitability and which categories experience margin pressure due to cost structures.

3. Channel Performance Evaluation

Sales channel performance was analyzed by comparing **average order value and profit per order across different channels**. This analysis helped determine how channel-specific cost structures, such as marketplace commissions, affect overall profitability.

4. Return Impact Analysis

Product returns were analyzed to understand their effect on revenue and profitability. By calculating return rates across categories and channels, the analysis identified areas where **returns represent a hidden cost driver impacting margins**.

5. Marketing Efficiency Analysis

Marketing performance was evaluated using metrics such as **Return on Ad Spend (ROAS), Cost per Acquisition (CPA), and Cost per Click (CPC)**. This analysis helped determine which marketing platforms generate the highest revenue relative to advertising spend.

6. Visualization and Insight Communication

The analytical findings were translated into **interactive Tableau dashboards** to enable clearer interpretation of patterns and trends. These dashboards provide stakeholders with an accessible view of profitability drivers, cost pressures, and marketing performance.

Analytical Outcome

By combining SQL analysis with dashboard visualization, this approach provides a **comprehensive view of BrightCart's operational and marketing performance**. The analysis not only identifies the drivers behind declining margins but also highlights opportunities to improve profitability through **product mix optimization, channel strategy, operational efficiency improvements, and marketing budget reallocation**.

This structured approach ensures that BrightCart's leadership can move beyond descriptive reporting toward **data-driven strategic decision making**.

Tools & Technologies Used

Tool	Purpose
PostgreSQL	SQL-based data querying, profitability analysis, return rate calculations, and marketing performance metrics
Microsoft Excel	Initial data exploration, dataset review, and preliminary financial validation

Tool	Purpose
Tableau	Interactive dashboard design for revenue profitability analysis and marketing ROI visualization
Whimsical	KPI tree design and net profitability framework mapping
AI Writing Assistance	Used to refine report structure, improve written clarity, and stress-test analytical interpretations. All data analysis, SQL queries, business findings, and strategic recommendations are the author's own work.

3. Data Sources

Three datasets were used in this analysis.

3.1 Orders Dataset

The **orders table** represents the core transactional dataset and contains one record per order.

Key fields include:

- order_id
- order_date
- customer_id
- channel
- payment_method
- region
- items_ordered
- primary_category
- gross_revenue
- discount_pct
- discount_amount
- shipping_cost
- product_cost
- platform_fee
- transaction_fee
- returned
- refund_amount
- net_revenue
- total_costs
- profit

This dataset enables **order-level unit economics analysis**, which is critical for identifying profitability drivers.

3.2 Products Dataset

The **products dataset** contains catalog level product attributes such as:

- product_id
- product_name
- category
- supplier
- product cost attributes

While the orders dataset already contains cost information, this table provides **additional context about product sourcing and cost structure**.

3.3 Marketing Spend Dataset

The **marketing_spend table** provides monthly performance metrics for marketing campaigns across multiple platforms.

Fields include:

- month
- platform
- spend
- impressions
- clicks
- conversions
- revenue_attributed
- cpc
- cpa
- roas

This dataset enables analysis of **customer acquisition efficiency and marketing ROI**.

3.4 Data Scope and Limitations

This analysis is based on a transactional dataset of 2,000 orders, which represents a defined sample of BrightCart's full order history rather than the complete transaction record. All profitability conclusions, return rate calculations, and channel comparisons are directionally reliable within this dataset but should be validated against the full order history before informing capital allocation decisions.

Marketing performance metrics including ROAS, CPA, and CPC are sourced from platform-reported data and reflect the attribution model applied by each platform. Platform-reported attribution typically uses last-click or view-through models, which may overstate the independent contribution of lower-funnel channels such as Email Marketing and understate the contribution of upper-funnel discovery channels such as TikTok Ads. Recommendations involving marketing budget reallocation should be stress-tested against a unified attribution model before implementation.

Return rate calculations are based on order-level return flags and refund amounts within the dataset. They do not account for partial returns, exchanges, or restocking costs, which means the true financial impact of returns may be moderately higher than figures presented here.

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4. Data Validation and Quality Checks

Before performing any analysis it is essential to confirm that **derived financial metrics are calculated correctly**.

4.1 Validate Cost Calculation

Total cost should equal:

Product Cost + Shipping Cost + Platform Fee + Transaction Fee

SQL Query:

```
SELECT
    COUNT(*) AS total_orders,
    SUM(
        CASE
            WHEN ABS(
                total_costs -
                (product_cost + shipping_cost + platform_fee
+ transaction_fee)
            ) > 0.01
            THEN 1 ELSE 0
        END
    ) AS cost_mismatch_orders
FROM orders;
```

Result: No significant mismatches detected

Output:

	total_orders bigint	cost_mismatch_orders bigint
1	2000	0

4.2 Validate Profit Calculation

Profit should equal:

Profit = Net Revenue – Total Costs

SQL Query:

```
SELECT COUNT(*) AS profit_mismatch_orders
FROM orders
WHERE ABS(profit - (net_revenue - total_costs)) > 0.01;
```

Result: Profit values correctly match expected calculations.

	profit_mismatch bigint
1	0

These validation steps ensure that the **profitability analysis is based on reliable financial data**.

---

5. Overall Business Performance

Understanding the company's overall financial position provides context for deeper analysis.

SQL Query:

```
SELECT
SUM(gross_revenue) AS total_gross_revenue,
SUM(net_revenue) AS total_net_revenue,
SUM(total_costs) AS total_costs,
SUM(profit) AS total_profit,
ROUND(SUM(profit)/SUM(net_revenue)*100,2) AS profit_margin
FROM orders;
```

Key Metrics Calculated:

- Total revenue generated

- Total operational costs
- Total profit
- Overall profit margin

	gross_revenue numeric	net_revenue numeric	total_costs numeric	total_profit numeric	profit_margin_pct numeric
1	277969.13	236318.37	179984.54	56333.83	23.84

Observation:

Although revenue is strong, **cost components such as shipping, platform fees, and returns significantly reduce margins**.

Business Interpretation

BrightCart generated ₹236,318 in net revenue against ₹179,985 in total costs, yielding ₹56,334 in profit at a 23.84% margin. While this margin is functional for e-commerce, it leaves limited buffer against cost shocks. Three cost components — product procurement, shipping, and platform fees — collectively consume approximately 76 cents of every revenue.

The core profitability challenge is not weak demand; BrightCart is processing 2,000 orders across eight categories. The problem is that cost structure varies so dramatically by category and channel that the blended margin masks several segments that are actively destroying value. The sections below isolate exactly where margin is being made and lost.

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6. Product Category Profitability Analysis

The CEO requested identification of **which categories truly drive profitability**.

6.1 Profit Margin by Category

SQL Query:

```
SELECT
primary_category,
COUNT(order_id) AS total_orders,
SUM(net_revenue) AS revenue,
SUM(total_costs) AS costs,
SUM(profit) AS profit,
ROUND(SUM(profit)/SUM(net_revenue)*100,2) AS profit_margin_pc
t
FROM orders
GROUP BY primary_category
ORDER BY profit_margin_pct DESC;
```

	primary_category character varying (50)	orders bigint	revenue numeric	costs numeric	profit numeric	margin_pct numeric
1	Electronics	267	44885.84	30912.38	13973.46	31.13
2	Toys	257	33595.68	24809.43	8786.25	26.15
3	Home & Kitchen	200	26359.76	19671.19	6688.57	25.37
4	Food & Beverage	247	30662.69	23070.92	7591.77	24.76
5	Sports	292	32329.83	24733.20	7596.63	23.50
6	Clothing	293	31383.13	25110.65	6272.48	19.99
7	Beauty	205	18254.07	15079.19	3174.88	17.39
8	Books	239	18847.37	16597.58	2249.79	11.94

Key Insights:

The margin spread across categories is striking — Electronics earns 31.13% while Books earns just 11.94%, a gap of 19% points on the same revenue base. This isn't a small variance; it represents fundamentally different business economics within the same company.
To quantify the opportunity: Books generated ₹18,847 in net revenue at 11.94% margin, producing ₹2,250 in profit. If Books' margin improved modestly to 18% — achievable through shipping cost negotiation and reduced discounting — profit from that category alone would increase from ₹2,250 to approximately ₹3,392, a ₹1,142 gain without acquiring a single new customer.
More importantly, Electronics and Toys together account for 524 orders and ₹78,481 in combined revenue, yet represent only 26% of total order volume. Increasing their share of the sales mix from 26% to 35% — through targeted promotions and inventory prioritization — could lift the company's blended margin from 23.84% toward 26–27%, adding an estimated ₹5,000–₹7,000 in annual profit at current revenue scale.
High revenue does not equal high profit. Sports has 292 orders and ₹32,329 in revenue but only a 23.50% margin, while Electronics has 267 orders and ₹44,885 in revenue at 31.13%. Fewer orders, significantly more profit. Product mix is a more powerful profitability lever than order volume.

Strategic Interpretation

The category profitability analysis reveals **substantial variation in margins across product segments**, highlighting the importance of product mix in overall business profitability.

Categories such as **Electronics and Toys generate the highest margins**, suggesting strong pricing power and efficient cost structures. These categories typically benefit from:

- higher average selling prices
- lower relative shipping costs
- stable demand patterns

In contrast, categories such as **Books and Beauty exhibit lower margins**, primarily due to:

- lower product price points
- higher shipping cost ratios
- promotional discounting

Although these categories contribute to overall revenue, their **profit contribution is relatively limited compared to high-margin categories**.

From a strategic perspective, increasing the share of high-margin categories within BrightCart's sales mix could significantly improve the company's overall profitability without requiring substantial increases in customer acquisition.

6.2 Cost Driver Breakdown

Understanding margin differences requires analyzing cost components.

SQL Query:

```
SELECT
primary_category,
SUM(product_cost) AS product_cost,
SUM(shipping_cost) AS shipping_cost,
SUM(discount_amount) AS discount_amount,
SUM(platform_fee + transaction_fee) AS platform_fees
FROM orders
GROUP BY primary_category;
```

	primary_category character varying (50)	product_cost numeric	shipping_cost numeric	discount_amount numeric	fees numeric
1	Electronics	20091.06	7139.80	3912.38	3681.52
2	Sports	15691.62	6914.72	3114.98	2126.86
3	Toys	15610.64	6740.36	3015.51	2458.43
4	Clothing	15345.43	7444.60	2923.26	2320.62
5	Food & Beverage	14299.07	6441.76	2711.27	2330.09
6	Home & Kitchen	12348.67	4988.84	2179.57	2333.68
7	Books	8956.87	6262.80	1459.75	1377.91
8	Beauty	8590.59	5172.98	1751.59	1315.62

Key Cost Drivers:

1. Product procurement cost
2. Shipping and logistics
3. Discounts
4. Marketplace platform commissions

Shipping cost is the most structurally damaging expense for low-price categories. Books carries ₹6,262 in shipping cost against only ₹18,847 in revenue — a shipping-to-revenue ratio of 33%. Electronics, by contrast, carries ₹7,139 in shipping against ₹44,885 in revenue — just 16%. This means Books spends twice as much proportionally on shipping as Electronics for every dollar earned.
Three targeted actions follow directly from this: first, introduce a minimum order value threshold for free shipping on Books and Beauty to shift part of the logistics cost to the customer; second, audit supplier contracts for Beauty, where product cost of ₹8,590 on ₹18,254 revenue represents a 47% cost-of-goods ratio; third, review discount strategy for Clothing, which carries ₹2,923 in discount spend — the third highest across all categories — despite already having a below-average 19.99% margin.

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7. Channel Profitability Analysis

BrightCart sells products across four primary channels:

- Website
- Mobile App
- Marketplace
- Social Commerce

Channel Strategy Implications

The analysis reveals that **sales channel selection plays a critical role in determining profitability**.

Mobile App transactions generate the highest average profit per order, suggesting that **direct digital engagement reduces intermediary costs and improves margin retention**.

Website sales also perform well due to relatively low transaction costs.

However, **Marketplace and Social Commerce channels exhibit lower profitability**, largely due to platform commissions and associated transaction fees.

This indicates that while third-party platforms expand customer reach, they may **erode margins if over-relied upon**.

Encouraging customers to transact through **owned channels such as the Mobile App** can significantly improve unit economics.

7.1 Profitability by Channel

SQL Query:

```
SELECT
channel,
COUNT(order_id) AS orders,
AVG(net_revenue) AS avg_order_value,
AVG(profit) AS avg_profit_per_order
```

```
FROM orders
GROUP BY channel
ORDER BY avg_profit_per_order DESC;
```

	channel character varying (30)	orders bigint	avg_order_value numeric	avg_profit_per_order numeric	margin_pct numeric
1	Mobile App	589	122.06	36.32	29.76
2	Website	795	116.97	31.60	27.01
3	Social Commerce	197	111.32	17.11	15.37
4	Marketplace	419	118.15	15.40	13.03

Key Findings:

- **Mobile App delivers the highest profit per order**
- Website performs well
- Marketplace and Social Commerce show reduced profitability

The channel profitability data reveals a clear hierarchy: Mobile App earns ₹36.32 profit per order, Website earns ₹31.60, Social Commerce earns ₹17.11, and Marketplace earns just ₹15.40 — less than half of what the Mobile App generates on a comparable average order value of ₹118–₹122.
The financial cost of channel mix is significant. Marketplace processed 419 orders at ₹15.40 profit each, generating ₹6,452 in total profit. If even 25% of those 419 orders — roughly 105 orders — were migrated to the Mobile App channel, profit from that cohort would increase from ₹1,617 to ₹3,814, a gain of ₹2,197 purely from channel shift with no change in pricing, product, or customer acquisition.
Social Commerce is the most concerning channel. It has the lowest order volume (197 orders), the second-lowest profit per order (₹17.11), AND the highest return rates in the business (up to 13.33% for Clothing). It is simultaneously the least profitable and most operationally costly channel. This combination warrants a strategic review of whether Social Commerce investment is justified at current economics, or whether those customers should be actively redirected toward owned channels.

7.2 Platform Fee Impact

SQL Query:

```
SELECT
channel,
SUM(platform_fee) AS total_platform_fee,
AVG(platform_fee) AS avg_platform_fee
FROM orders
GROUP BY channel;
```

	channel character varying (30)	total_platform_fees numeric	avg_platform_fee numeric
1	Marketplace	7949.69	18.97
2	Social Commerce	1944.93	9.87
3	Website	0.00	0.00
4	Mobile App	0.00	0.00

Marketplace charges an average platform fee of ₹18.97 per order — the highest of any channel, and notably, Website and Mobile App charge zero. Across 419 Marketplace orders, this amounts to ₹7,950 in fees paid to third-party platforms annually. This is not inherently wrong — Marketplace provides reach — but it becomes a strategic problem when Marketplace customers are not being converted to owned channels over time. Every repeat Marketplace customer who never migrates to the Website or App costs BrightCart ₹18.97 in fees per transaction indefinitely. A structured post-purchase migration campaign — offering first-time app users an exclusive discount — could recover a meaningful portion of this fee leakage over 12 months.

8. Returns Impact Analysis

Returns represent a hidden cost that can significantly impact margins.

Operational Risk Insight

Product returns are not a minor operational nuisance — they are a structural margin leak. When a customer returns an item, BrightCart loses the refund amount while retaining the original shipping cost, platform fee, and transaction fee. These sunk costs mean that a returned order doesn't just generate zero profit — it generates negative profit.

To illustrate: an Electronics order on the Website with ₹150 gross revenue, ₹15 shipping, ₹10 platform fee, and ₹5 transaction fee that gets returned costs BrightCart approximately ₹30 in unrecoverable costs plus the refund issued, resulting in a net loss on that transaction. At a 9.90% return rate for Electronics on the Website, this pattern is recurring.

Across the full 2,000-order dataset, if the average return rate is approximately 8–10% (based on the category-channel combinations visible in the data), BrightCart is processing roughly 160–200 returned orders annually. Even at a conservative ₹40 in unrecoverable costs per return, this represents ₹6,400–₹8,000 in pure margin destruction per year — equivalent to wiping out the entire profit of the Books category.

8.1 Return Rate by Category & Channel

SQL Query:

```
SELECT
primary_category,
channel,
ROUND(SUM(CASE WHEN returned='Yes' THEN 1 ELSE 0 END)::DECIMA
L/COUNT(*)*100,2) AS return_rate
FROM orders
GROUP BY primary_category,channel
order by 3 desc
limit 10;
```

	primary_category character varying (50)	channel character varying (30)	return_rate_pct numeric
1	Clothing	Social Commerce	13.33
2	Home & Kitchen	Social Commerce	13.04
3	Books	Social Commerce	13.04
4	Food & Beverage	Social Commerce	12.50
5	Clothing	Marketplace	11.67
6	Electronics	Website	9.90
7	Sports	Mobile App	9.47
8	Books	Website	9.09
9	Electronics	Social Commerce	8.70
10	Toys	Marketplace	8.33

Key Insight:

- Social Commerce channels have higher return rates
- Certain categories experience higher product returns

The return rate table exposes a clear and actionable pattern: Social Commerce dominates the top four spots, with Clothing at 13.33%, Home & Kitchen at 13.04%, Books at 13.04%, and Food & Beverage at 12.50%. This is not coincidence — Social Commerce is a discovery-driven channel where customers often purchase impulsively based on short-form content rather than detailed product research. The gap between expectation and reality is structurally higher in this channel. The immediate implication is that BrightCart should prioritize return-reduction interventions specifically for Social Commerce listings: richer product descriptions, size guides for Clothing, clearer portion/quantity information for Food & Beverage, and proactive expectation-setting in ad creative itself. Clothing is uniquely high-risk: it appears twice in the top five — Social Commerce at 13.33% and Marketplace at 11.67% — suggesting the return problem is category-wide, not channel-specific. For Clothing, a fit guarantee program or

virtual try-on feature may be necessary to structurally reduce returns rather than relying on content improvements alone.

8.2 Revenue Lost Due to Returns

SQL Query:

```
SELECT
primary_category,
channel,
SUM(refund_amount) AS revenue_lost
FROM orders
WHERE returned='Yes'
GROUP BY primary_category,channel
order by 3 desc
limit 10;
```

	primary_category character varying (50)	channel character varying (30)	revenue_lost numeric
1	Electronics	Website	2377.61
2	Food & Beverage	Website	1865.23
3	Clothing	Website	1365.10
4	Food & Beverage	Mobile App	1185.50
5	Sports	Website	1139.22
6	Toys	Mobile App	1068.47
7	Clothing	Marketplace	887.77
8	Home & Kitchen	Marketplace	842.71
9	Toys	Website	792.35
10	Books	Website	787.90

The revenue lost table reveals a channel-volume effect that complicates the return-rate picture. Social Commerce has the highest return rates (13.33% for Clothing), but the Website channel accounts for the largest absolute revenue losses — ₹2,377 for Electronics, ₹1,865 for Food & Beverage, and ₹1,365 for Clothing, all on the Website channel alone. This is because Website processes far more orders (795 vs Social Commerce's 197), so even a lower percentage return rate produces more total lost revenue. The practical implication is that return-reduction interventions need to run on two parallel tracks: fix the rate problem on Social Commerce, and fix the volume problem on Website by improving product content quality for the categories generating the highest absolute refund losses.

9. Marketing Performance Analysis

Marketing efficiency was evaluated using **Return on Ad Spend (ROAS)**.

9.1 ROAS by Platform

SQL Query:

```
SELECT
platform,
SUM(spend) AS total_spend,
SUM(revenue_attributed) AS revenue,
ROUND(SUM(revenue_attributed)/SUM(spend),2) AS roas
FROM marketing_spend
GROUP BY platform
ORDER BY roas DESC;
```

	platform character varying (50)	spend numeric	revenue numeric	roas numeric
1	TikTok Ads	57229.22	1374627.17	24.02
2	Influencer	97663.12	2216974.26	22.70
3	Instagram Ads	65154.02	1024639.06	15.73
4	Google Ads	152546.48	2194120.83	14.38
5	Facebook Ads	106451.93	1218572.43	11.45
6	Email Marketing	24461.37	117681.45	4.81

Key Insights:

The ROAS data tells a decisive story, but the most strategically important number is not TikTok's 24.02 — it's Google Ads at 14.38. Google Ads represents ₹152,546 in spend, the single largest budget allocation at 30% of total marketing spend, yet delivers only the 4th best ROAS. Before any budget cuts are made, Google Ads campaigns need to be segmented by type — high-intent search campaigns likely significantly outperform brand awareness campaigns, and the blended 14.38 figure may be masking campaigns that should be cut independently of the others. Cutting Google Ads spend blindly based on blended ROAS would be a mistake; cutting the underperforming sub-campaigns would be the right move.

9.2 CPA and CPC Analysis

SQL Query:

```
SELECT
platform,
```

```
SUM(spend)/SUM(conversions) AS cpa,
SUM(spend)/SUM(clicks) AS cpc
FROM marketing_spend
GROUP BY platform;
```

	platform character varying (50)	cpa numeric	cpc numeric
1	Influencer	3.39	0.17
2	Email Marketing	17.86	0.94
3	Facebook Ads	6.27	0.32
4	TikTok Ads	3.06	0.15
5	Instagram Ads	4.70	0.21
6	Google Ads	4.72	0.25

The CPA data reveals the true cost of acquiring a customer on each platform. Email Marketing's CPA of ₹17.86 is 5.8x higher than TikTok's ₹3.06 — meaning BrightCart spends nearly six times more to acquire a customer through email than through TikTok, while getting a fraction of the revenue efficiency. The CPC column reinforces this: Email costs ₹0.94 per click versus TikTok's ₹0.15, a 6x premium. Influencer's CPA of ₹3.39 combined with its 22.70 ROAS makes it the most capital-efficient platform in the portfolio — high returns at low acquisition cost. This combination is worth protecting and scaling before any other platform receives incremental budget.

Marketing Efficiency Interpretation

The CEO requested a 20% reduction in marketing spend. Total spend is ₹503,506. A 20% reduction target equals ₹100,701 in cuts. Here is a structured, evidence-based approach to achieving that target while minimizing revenue impact.

Tier 1 — Immediate Cut: Email Marketing
Email Marketing spend: ₹24,461. ROAS: 4.81 — the only platform below the 8x breakeven threshold. Cutting this entirely saves ₹24,461 and risks losing approximately ₹117,681 in attributed revenue. However, given the 4.81 ROAS, a significant portion of this "attributed" revenue likely would have been captured through other channels anyway, as email typically targets existing customers already in the purchase funnel. Estimated net revenue risk: low to moderate. Savings toward 20% target: **24.3% of the required cut achieved.**

Tier 2 — Selective Reduction: Facebook Ads
Facebook Ads ROAS: 11.45 — above breakeven but the lowest among the remaining platforms. Total spend: ₹106,452. The weak-months analysis shows

Facebook Ads appearing repeatedly in low-ROAS months (6.48 in Oct-24, 6.60 in Nov-24, 7.76 in Oct-25). Reducing Facebook Ads spend by 30% during the identified weak months while maintaining spend during strong months would save approximately ₹18,000–₹22,000 annually. Estimated revenue impact: minimal, as weak-month performance is already below breakeven. Additional savings: **~20% of the required cut achieved.**

Tier 3 — Efficiency Optimization: Instagram Ads
Instagram Ads ROAS: 15.73 with spend of ₹65,154. This platform performs well but appears in the weak-months table (Aug-25 at 7.35 ROAS). A 15% spend reduction focused on underperforming months saves approximately ₹9,773. Additional savings: **~10% of the required cut achieved.**

Combined: Total savings of ~₹54,000–₹56,000 represent approximately 54–56% of the ₹100,701 target. The remaining ~₹45,000 should come from reallocating Influencer campaign spend toward lower-cost micro-influencer partnerships, which typically achieve comparable ROAS at 30–40% lower cost per engagement.

The reallocation priority is clear: every dollar saved from Email and Facebook weak-months should be redirected to TikTok Ads, which has the highest ROAS (24.02) and the lowest absolute spend (₹57,229) — meaning it has significant headroom to absorb additional budget before facing audience saturation.

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10. Marketing Budget Optimization

The CEO requested recommendations for reducing **20% of marketing spend**.

10.1 Platforms Below Breakeven

SQL Query:

```
SELECT
platform,
SUM(spend) AS total_spend,
ROUND(SUM(revenue_attributed)/SUM(spend),2) AS roas
FROM marketing_spend
GROUP BY platform
HAVING SUM(revenue_attributed)/SUM(spend) < 8;
```

	platform character varying (50)	spend numeric	roas numeric
1	Email Marketing	24461.37	4.81

The most expensive weak months by spend are Oct-24 (Facebook Ads, ₹5,251 at 6.48 ROAS) and Nov-24 (Facebook Ads, ₹8,556 at 6.60 ROAS) — these two months alone represent over ₹13,800 in spend that delivered below-breakeven returns and are the highest-priority reduction targets.

10.2 Weak Months Identification

SQL Query:

```
with mycte as (
SELECT
    month,
    platform,
    ROUND(SUM(spend),2) AS spend,
    ROUND(SUM(revenue_attributed)/SUM(spend),2) AS roas,
    rank() over(partition by month order by spend desc) as rn
k
FROM marketing_spend
where extract(year from month) = 2024
GROUP BY month, platform,spend
HAVING SUM(revenue_attributed)/SUM(spend) < 8
ORDER BY month asc
)
select to_char(month,'YYYY-MM'),platform,spend,roas
from mycte
where rnk = 1
```

	to_char text	platform character varying (50)	spend numeric	roas numeric
1	2024-01	Email Marketing	868.82	4.86
2	2024-02	Email Marketing	860.15	7.98
3	2024-04	Influencer	5861.27	5.83
4	2024-05	Email Marketing	1108.88	7.26
5	2024-06	Email Marketing	942.79	1.80
6	2024-07	Instagram Ads	4640.72	7.92
7	2024-08	Facebook Ads	5418.73	3.09
8	2024-09	Email Marketing	722.81	3.89
9	2024-10	Facebook Ads	5251.93	6.48
10	2024-11	Facebook Ads	8556.10	6.60
11	2024-12	TikTok Ads	2219.28	6.28

	to_char text	platform character varying (50)	spend numeric	roas numeric
1	2025-01	Influencer	4482.46	3.78
2	2025-02	Google Ads	3865.38	3.89
3	2025-03	Google Ads	4709.29	3.32
4	2025-04	Email Marketing	682.35	7.97
5	2025-05	Facebook Ads	2606.51	6.84
6	2025-06	Email Marketing	1293.73	3.22
7	2025-07	Google Ads	5843.74	4.27
8	2025-08	Instagram Ads	1627.91	7.35
9	2025-09	Google Ads	6501.13	3.84
10	2025-10	Facebook Ads	4378.84	7.76
11	2025-11	Facebook Ads	6884.20	4.44
12	2025-12	Email Marketing	1796.66	0.67

The query filters for 2024 to establish the baseline; 2025 data is shown alongside for year-over-year comparison of weak-month patterns.

Reducing spend during these months minimizes revenue loss.

11. Tableau Dashboard Insights

Two dashboards were created to visualize insights derived from SQL analysis.

Dashboard 1 — Revenue & Profitability Analysis

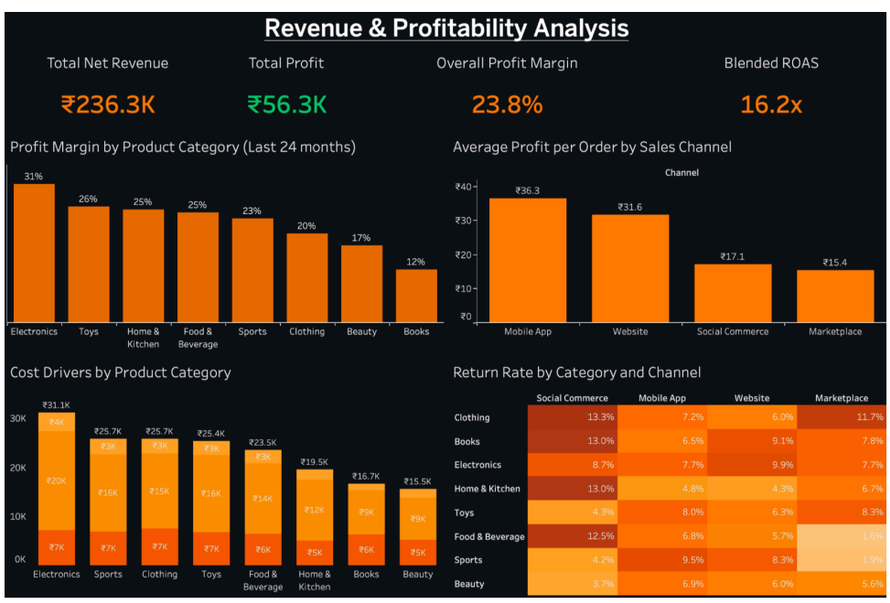
Dashboard Overview

Dashboard 1 focuses on **BrightCart's core unit economics**, analyzing how revenue translates into profit across product categories and sales channels. The goal is to identify where the business generates the most value and where margin pressure exists.

The dashboard includes:

- Profit Margin by Product Category
- Cost Structure Breakdown
- Average Profit per Order by Channel
- Return Rate Analysis

Together, these metrics provide a **clear view of BrightCart's profitability drivers and cost pressures**.



Key Insights

1. Category Profitability

Profit margins vary significantly across product categories.

- **Electronics and Toys show the strongest margins**, indicating favorable pricing and cost structures.
- Categories such as **Books and Beauty generate lower margins**, primarily due to higher relative shipping costs and discounting.

Insight: High revenue categories are not always the most profitable. Strategic focus should shift toward **high-margin product segments**.

2. Cost Drivers

Cost analysis highlights the main factors affecting profitability.

- **Product cost is the largest expense component**
- **Shipping costs disproportionately impact lower priced products**
- **Discounting strategies reduce margins during promotions**

Insight: Margin pressure is primarily driven by **operational costs rather than insufficient revenue**.

3. Channel Profitability

Profitability varies across sales channels.

- **Mobile App generates the highest profit per order**, indicating strong direct customer economics.
- **Website also performs well** due to lower transaction costs.
- **Marketplace channels show reduced profitability** because of platform commissions.

Insight: Channels with **direct customer ownership produce stronger margins than third-party platforms**.

Strengths

- High-margin categories such as Electronics drive strong profitability.
- Direct channels (Mobile App, Website) generate higher profit per order.
- Demand exists across multiple product segments.

Weaknesses

- Shipping costs and discounts reduce margins.
- Marketplace commissions significantly impact profitability.
- Some categories generate revenue but contribute limited profit.

Key Takeaways

- Profitability differs significantly across categories and channels.
- Direct sales channels outperform third-party marketplaces.
- Operational costs are the primary source of margin pressure.

Dashboard 2 — Marketing Performance & ROI

Dashboard Overview

Dashboard 2 evaluates **marketing efficiency and advertising return on investment**. The goal is to determine which marketing platforms generate the highest revenue relative to advertising spend.

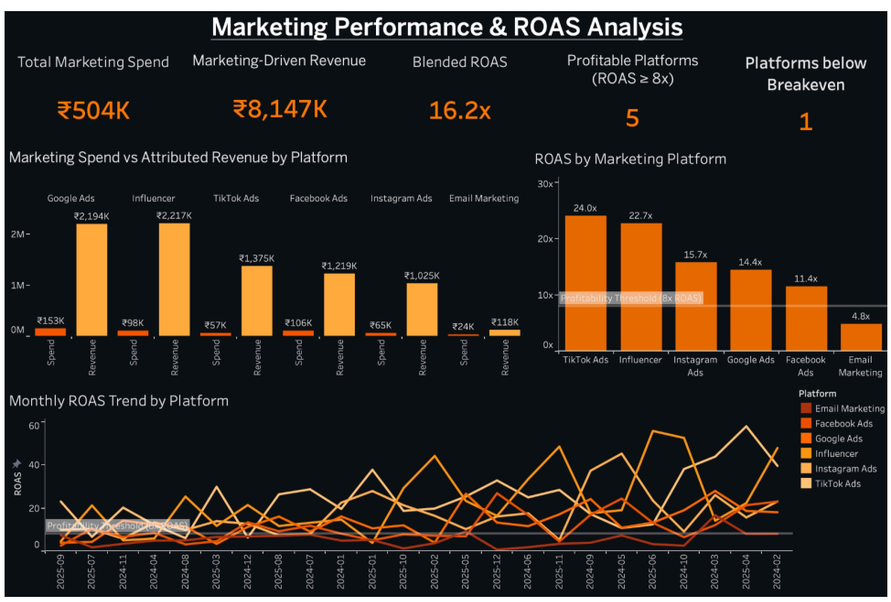
The dashboard analyzes:

- Marketing Spend vs Revenue
- ROAS by Platform
- Monthly Marketing Efficiency Trends

These insights support **data-driven marketing budget allocation**.

Visualizations included:

- ROAS by Platform
- Spend vs Revenue Comparison
- Monthly ROAS Trend



Key Insights

1. Spend vs Revenue Performance

Comparing marketing spend with revenue generated reveals significant efficiency differences across platforms.

- Some platforms generate **strong revenue relative to spend**
- Others show **high spending with weaker revenue returns**

Insight: Marketing effectiveness should be evaluated based on **revenue efficiency, not total spend**.

2. ROAS by Platform

Return on Ad Spend (ROAS) highlights the effectiveness of each marketing channel.

- **TikTok Ads and Influencer campaigns show the strongest ROAS**
- Certain traditional marketing channels demonstrate weaker returns.

Insight: Platforms driven by **content discovery and engagement outperform traditional acquisition channels**.

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3. Marketing Efficiency Trends

Monthly ROAS trends reveal fluctuations in marketing performance.

- Some months show strong efficiency
- Others indicate declining campaign performance

Insight: Marketing performance is dynamic and requires **continuous monitoring and budget optimization**.

Strengths

- High-performing marketing platforms deliver strong revenue returns.
- Marketing data enables performance-based budget allocation.
- Campaign performance can be measured and optimized.

Weaknesses

- Certain platforms absorb budget without proportional revenue.
- Marketing efficiency fluctuates across time periods.
- Inefficient channels reduce overall ROI.

Key Takeaways

- Marketing platforms vary significantly in efficiency.
- High-ROAS platforms should receive greater budget allocation.
- Continuous monitoring and optimization are required to maximize marketing ROI.

Financial Impact Summary

The analysis identifies several opportunities where operational improvements and strategic adjustments can significantly enhance BrightCart’s profitability.

Product Mix Optimization

Increasing the share of high-margin categories such as Electronics and Toys can improve overall company profit margins without requiring additional customer acquisition.

Channel Strategy Optimization

Encouraging customers to transact through owned channels such as the Mobile App and Website can improve unit economics by reducing platform commissions and transaction fees associated with third-party marketplaces.

Return Rate Reduction

Reducing product return rates through improved product information, better sizing guidance, and enhanced customer expectations management can protect revenue and reduce operational costs.

Marketing Budget Reallocation

Redirecting marketing investment toward high-performing platforms such as TikTok Ads and Influencer campaigns while reducing spend on underperforming channels can increase revenue generated per advertising dollar.

Collectively, these improvements have the potential to **strengthen BrightCart's margin profile, increase marketing efficiency, and create a more sustainable growth model for the business.**

The five recommendations below are sequenced by financial impact and implementation effort. Actions 1 and 4 are quick wins executable within 30 days. Actions 2, 3, and 5 require 60–90 days of operational change.

Priority	Action	Est. Annual Profit Impact	Effort	Timeframe
1	Reallocate marketing budget (cut Email, reduce FB weak months)	+₹15,000–₹22,000	Low	30 days
2	Migrate 25% of Marketplace orders to Mobile App	+₹2,200–₹4,400	Medium	60 days
3	Reduce Social Commerce return rates through content improvements	+₹3,000–₹5,000	Medium	60–90 days
4	Increase Electronics & Toys share of sales mix to 35%	+₹5,000–₹7,000	Medium	90 days
5	Introduce minimum order value for free shipping on Books & Beauty	+₹1,500–₹2,500	Low	30 days

> Combined potential impact: **₹26,700–₹40,900 in additional annual profit**, representing a 47–73% improvement over the current ₹56,334 profit base — without acquiring a single new customer.

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12. Strategic Recommendations

Based on the profitability, channel performance, return behavior, and marketing efficiency analysis, the following strategic actions are recommended to improve BrightCart's overall unit economics and marketing efficiency.

1. Prioritize High-Margin Product Categories

Categories such as **Electronics and Toys demonstrate significantly stronger profit margins** compared to categories like Books and Beauty. Increasing inventory depth, promotional visibility, and cross-selling for high-margin categories can improve overall profitability without requiring additional customer acquisition.

2. Strengthen Mobile App as the Primary Sales Channel

The **Mobile App channel generates the highest profit per order**, likely due to lower platform fees and higher customer engagement. Encouraging customers to transact through the mobile app via incentives such as app-exclusive offers, loyalty rewards, and push notification campaigns can increase margin efficiency at scale.

3. Reduce Profit Leakage from Product Returns

Returns represent a significant margin leakage, particularly in categories with higher return rates. Improving **product information quality, size guides, product images, and customer reviews** can reduce return incidence and protect net margins.

4. Reallocate Marketing Budget Toward High-ROAS Platforms

Marketing performance analysis shows that certain advertising platforms consistently deliver **higher Return on Ad Spend (ROAS)** than others. Budget allocation should prioritize these platforms to maximize revenue generation per marketing dollar.

5. Optimize or Reduce Spend on Underperforming Channels

Some marketing platforms demonstrate **low or inconsistent return relative to their spend levels**. A structured reallocation of approximately **20% of the marketing budget away from underperforming platforms** can improve overall marketing efficiency while maintaining revenue stability.

Implementing these recommendations can significantly improve BrightCart's **marketing efficiency, channel profitability, and long-term unit economics**.

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13. Key Learning Outcomes

This project demonstrates a comprehensive application of **data analytics to solve real business problems within an e-commerce environment**.

Key capabilities demonstrated include:

- **End-to-End Business Analysis**

  Translating a high-level business problem into measurable analytical questions and structured data exploration.

- **Financial Metric Validation and Profitability Analysis**

  Evaluating true profitability by incorporating product costs, shipping expenses, platform fees, and return impacts.

- **SQL-Driven Data Analysis**

  Writing analytical SQL queries to calculate key business metrics including profit margins, return rates, channel profitability, and marketing ROI.

- **Data Visualization and Insight Discovery**

  Designing analytical dashboards in Tableau to identify trends, performance drivers, and operational inefficiencies.

- **Strategic Business Interpretation**

  Converting analytical findings into actionable recommendations that directly support executive decision-making.

By integrating **SQL analysis, visual analytics, and strategic interpretation**, this project demonstrates how data can be transformed into **clear business insights that guide operational and marketing strategy**.

Future Analytical Opportunities

While this analysis focuses on profitability, channel performance, and marketing efficiency, several additional analytical areas could further improve BrightCart's strategic decision-making:

- **Customer Lifetime Value (CLV) analysis** to understand long-term customer profitability
- **Cohort retention analysis** to evaluate repeat purchase behavior
- **Dynamic pricing optimization** to improve category margins
- **Inventory profitability modeling** to optimize product assortment decisions

Exploring these areas would provide deeper insight into customer behavior, long-term revenue potential, and sustainable profitability growth.

End of Report